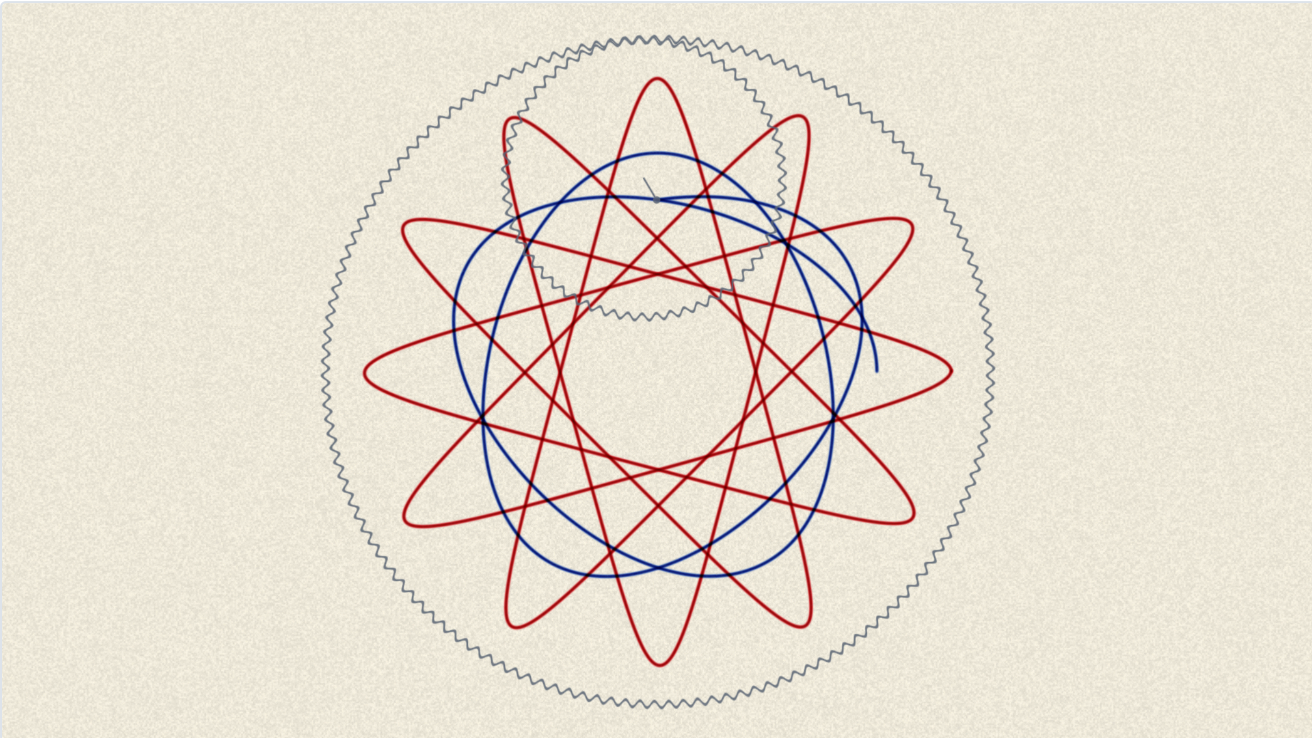
Cogwheel user guide

Cogwheel is **a Spirograph**, as two plugins for Resolume Arena and Avenue — and again as an OpenFX bundle for Resolve, Vegas, Nuke and Natron.

A toothed ring is pinned to the paper. A smaller toothed wheel rolls around inside it, in mesh, and a pen dropped through one of the wheel's holes is carried along for the ride. **Nothing here evaluates a curve.** Set the two tooth counts and the figure follows.



> **Before you rely on this:** the central claims are measured rather than asserted. The pen is walked for the full number of turns the tooth counts predict and the test fails if it does not come home; a second fails unless two pens crossing transmit the *product* of what each transmits alone, which is what makes ink subtractive rather than additive. Both probe the same code that ships, and a control sweep fails if any parameter turns out to do nothing.
>
> **Both plugins have been loaded and run in Resolume Arena 7.27.1, on macOS and on Windows** — they register with the right names and categories, expose all 48 controls with nothing truncated, render, and hold a factory preset through live rendering. **DaVinci Resolve Studio 21.0.2.4 lists the OpenFX build**, and it renders correctly in an independent OFX host. What is still unconfirmed is narrower: the OpenFX build has not been driven over a clip inside Resolve, so its CPU renderer is proven to run and not yet proven to match the GPU one pixel for pixel.
>
> Released at **v0.2.0**.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

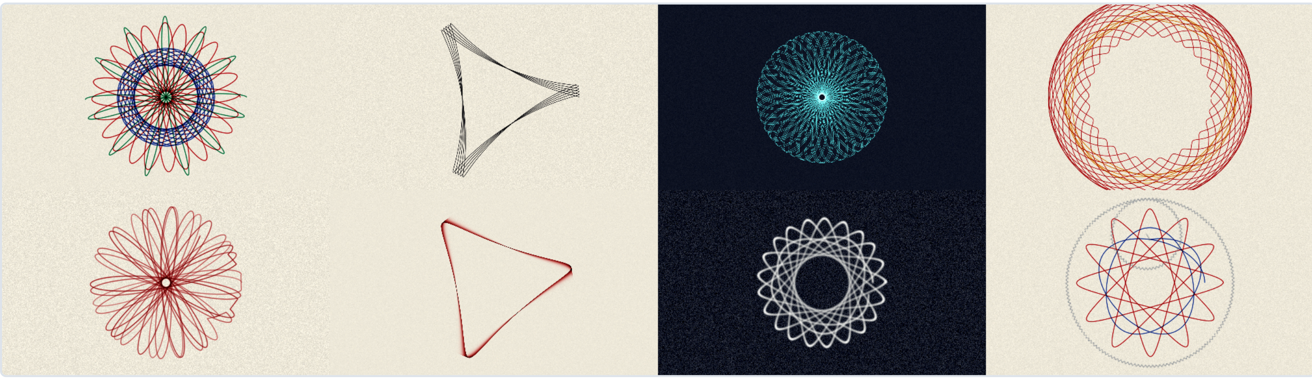
Two plugins

- **Cogwheel** (a source) draws on its own sheet of paper.
- **Cogwheel Ink** (an effect) can use the clip below as the paper the pen draws on, as the ink the pen picks up, or both.

Both declare the same parameters, so a composition can be moved between them.

Start from a preset

The **Preset** dropdown is at the bottom of the list. Eight of them, and each is a recognisable *drawing* rather than a set of slider positions — which is only possible because the gear train is two integers.



Preset	What it is
Classic Rosette	96 and 52, four pens through four holes. The one everybody made first.
Three Lobes	96 and 32 mesh three to one, so the figure closes in a single turn. The pen never lifts, and a mesh half a tooth out of true walks it round the sheet.
Dense Web	105 and 64 share no factor at all: sixty-four turns and a hundred and five lobes, printed as a negative.
Flower	Outside the ring instead of inside — the petal shape rather than the rosette.
Slipping Gear	The ruined drawing, on purpose. A mesh well out of true and a three-tooth jump every few turns.
Wet Trail	Not a drawing: a trail. A fibre tip against a two-second fade.
Chalkboard	Graphite, a heavy tooth and a wide soft nib, printed as a negative.
Show the Gears	The machine on screen — ring, wheel, arm and pen — cranked slowly.

Keeping a look of your own

The presets are fixed — Resolume gives a plugin no way to add an entry to that dropdown while it is running, or to remember one between sessions. So there is no "save preset" button, and there cannot be one.

What there is, new in v0.2.0, is **Export XML**, next to the dropdown. It writes every control's current value to a timestamped file:

```
~/Documents/cogwheel/cogwheel-20260901-152822.xml
```

Each line carries both the number the plugin stores and the readable version of it, so the file is worth reading as well as keeping:

```
<parameter id="1" name="Ring Teeth" type="integer" value="96.000000" display="96t - 24 lobes"/>
<parameter id="8" name="Speed" type="standard" value="0.737000" display="1.63 turns/s"/>
```

It is a record, not a recall — nothing loads it back. Use it to keep a look you like, to send one to somebody, or to read the numbers off and type them in again. The full path is written to the diagnostics log each time, because sixteen characters of parameter display is not room for one.

Moving any control a preset covers drops the dropdown back to **Custom**. That is the preset letting go, not an error.

The two numbers that matter

Ring Teeth and **Wheel Teeth** are the whole machine. Because gears mesh, the wheel cannot roll by anything other than a whole tooth, so everything about the figure follows from those two integers:

Ring	Wheel	Lobes	Turns to close
96	32	3	1
96	52	24	13
96	31	96	31

Drag either control and Resolume shows you the answer before you let go — the Ring reports how many lobes you are about to get, and the Wheel reports how many turns it will take to close.

Snap to Set restricts both to tooth counts a real Spirograph set carries. Leave it on unless you are hunting for something specific; the figures people recognise are the ones those particular integers make.

Pen Hole is which hole in the wheel the pen sits in. Near the rim gives long thin lobes; near the axle gives something close to a circle. **Snap to Holes** restricts it to the twelve modelled positions, which is what a real wheel offers.

Mesh decides whether the wheel runs round the inside of the ring (a rosette) or the outside (a petal shape).

Why a drawing stops, and what to do about it

A closed figure retraces its own line for ever. That is not a bug and it is not something tuning fixes — it is what a Spirograph *is*. A drawing is finished when it is finished, and about ten seconds later nothing on screen is moving.

There are three ways out, and all three are things a person at the table actually does:

- **Creep** — the mesh is not quite true. A fraction of a tooth per turn precesses the figure so it never quite lands on itself, and the drawing keeps growing. This is on by default.

- **Skip** — a jumped tooth. Discrete, startling, and the classic way a real drawing is ruined. **Skip Size** is how far it jumps.
- **Layers** — when a figure closes, lift the pen, move to another hole, change the pen and draw the next one on top. That is how the multicoloured Spirograph drawing everybody remembers is actually made. **On Closing** decides what moves; **Wipe Sheet** starts a fresh sheet when the whole stack is done.

If the picture has gone still, one of those three is off.

The pen

Pen Type is a real distinction between two real classes of pen, not a look.

- A **ballpoint**'s ball rolls, so it lays down ink per unit of *distance*: a line is the same darkness however fast your hand moved. That is what came in the box, and it is the default.
- A **fibre tip** feeds by capillary action, per unit of *time*, so it blooms wherever the pen slows down — which at a cusp is a great deal.

Flow is how dark the line is. It is multiplied by the nib's width internally, so widening the nib does not lighten the line — a broader pen delivers proportionally more ink, which is what makes it broader rather than blurrier.

Nib is the line's width, as a fraction of the sheet, so it is the same line at every output resolution. **Pressure** is how much a pressed nib spreads.

Pens chooses what the layers are drawn with — *Four Pens* is the set that came in the box. *Ink Colour* uses the single **Ink** swatch instead.

Ink is subtractive, and what follows from it

The sheet accumulates optical density, and what you see is the paper through it. A second pen crossing the first therefore gives the *product* of the two transmissions — the muddy near-black it is on the table — rather than the bright sum an additive renderer would produce. A pen that lingers lays down more.

Two consequences worth knowing:

- **Changing the paper colour or the palette does not spoil what is already drawn.** The sheet never held a colour to be wrong about.
- **Ink can only ever darken paper.** A pale line on a dark ground is not something the machine can make. **Print → Negative** is the honest route to it: a drawing, photographed and printed the other way up.

Grain is how much of the paper's tooth you can see. **Tooth** is a different question — how unevenly the sheet *takes* ink. Smooth board takes ink evenly and shows no grain; cartridge does both.

Fade is the one control in the plugin that is not something the machine can do. A drawing does not fade. It is here because a VJ needs the sheet to clear, and at zero it does not fade at all, which is what paper does.

Cranking, and syncing to the music

Speed is turns of the ring per second.

> **Renamed in v0.2.0.** This control was called **Crank** in v0.1.0. Every other plugin in the range calls it Speed, so it now does too. Two things follow, and both bite silently: a composition saved against v0.1.0 loses whatever Crank was set to and opens at the default, and any OSC or MIDI mapping pointing at "Crank" stops arriving. Re-point the mapping and re-save the composition once, and that is the end of it.

Sync overrides it: choose 1, 2, 4 or 8 bars and the figure is given that many bars to complete, whatever this particular train's turn count is. So a thirteen-turn figure and a one-turn figure both land on the bar line together. That is the only sensible reading of "one figure per phrase".

New Sheet throws the drawing away and starts again. It sits at the very top of the parameter list, above every group, so it is always in reach — clearing the paper is the one thing you want mid-show and it used to be folded away inside the Crank group.

Detail is how finely the pen path is walked — 360, 1440 or 5760 steps a turn. It is a cost dial with a visible symptom: at *Draft* a big figure is faintly polygonal. It does **not** change how much ink goes on the sheet.

Show Gears puts the ring, the wheel, the arm and the pen on screen, with the real tooth counts, so they can be counted. It is the quickest way to explain the plugin to somebody standing next to you.

In OpenFX hosts

Copy `Cogwheel.ofx.bundle` to:

- **macOS** — `/Library/OFX/Plugins`
- **Windows** — `C:\Program Files\Common Files\OFX\Plugins`
- **Linux** — `/usr/OFX/Plugins`

⚠️ **On macOS it must be `/Library/OFX/Plugins`, which needs an administrator.** Putting it in `~/Library/OFX/Plugins` is **silently ignored** — the plugin simply never appears, with nothing logged and no error, which looks exactly like a broken plugin.

Two things behave differently there, both stated rather than hidden:

- **The OpenFX build renders on the CPU.** There is no OpenGL context to be had in an OFX host, so it is built for offline rendering rather than for live use.
- **A drawing is history, so it replays from the beginning.** A linear render costs nothing per frame beyond that frame's own strokes. Scrubbing *backwards* through a long drawing is slow, and gets slower the further in you are. There is no warm-up window that would fix this: one would produce a different drawing, missing everything laid down before the window opened. Render forwards.

Sync works off a fixed 120 bpm in OpenFX, because OFX carries no transport tempo.

If it looks wrong

The picture has stopped moving. The figure has closed. Turn up **Creep**, raise **Layers**, or add some **Skip**. See "Why a drawing stops" above.

Everything is solid black. Flow is too high for the crank rate. A dense figure crosses itself constantly and every crossing darkens; *Dense Web* runs at a deliberately light flow for exactly this reason.

The drawing is off the edge of the frame. Check **Size** and **Centre**. Size is the ring's radius as a fraction of the frame height, so at 1.0 the ring exactly fills it.

The lines look polygonal. Detail is on *Draft*. Move it to *Normal*.

Nothing appears at all in Resolume. Check `~/Library/Logs/cogwheel/` (macOS) or `%LOCALAPPDATA%\cogwheel\logs\` (Windows). A shader that will not compile, or a sheet that would not allocate, both look like "it does nothing" from outside and both say so there.

A control does nothing. Some are conditional: **Skip Size** needs **Skip Chance** above zero, **On Closing** and **Pens** need **Layers** above one, **Ink** only applies when **Pens** is set to *Ink Colour*, and **Ink from Clip** / **Paper from Clip** exist only on the effect.
